CS 412: Spring '26
Introduction To Data Mining

Assignment 6

(Due Wednesday, April 29, 23:59)

- The Exam is due on **Wednesday, April 29, 2026, at 23:59**. We are using Gradescope for all written assessments. The instructor sent an invitation to join Gradescope on Saturday, January 14th, 2024. Please get in touch with the instructor if you have issues accessing Gradescope. Contact the TAs if you have any technical difficulties while submitting the exam. Please do NOT email a copy of your solution. We will NOT accept late submissions (without a reasonable justification).
- Please use Campuswire if you have questions about the exam. Make sure to tag your post appropriately. Also, scroll through previous posts to make sure that your query was not answered previously.
- Make sure that your answers are clear and concise. The exam should be submitted as a single PDF file. You are expected to typeset the solutions (using typesetting software such as LaTeX). Handwritten solutions *may not* be graded.
- For a particular question, you will NOT receive full credit if you only provide the final result. Please show all necessary steps and details, including any formulae you use. Clearly describe the variables utilized within the formulae.

  Consider the following two examples.

Example 1 **Q:** Given a dataset $\mathcal{X} = \{3.1, 4.2, -1\}$, compute the mean.
A: For any set of n numbers $\mathcal{X} = \{x_1, \ldots, x_n\}$, the mean can be computed as $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$. For the given dataset $\mathcal{X}$, the mean is $\mu = \frac{3.1+4.2-1}{3} = 2.1$

Example 2 **Q:** A coin claimed to be unbiased has been tossed 100 times, with 54 heads and 46 tails. What is the χ^2 statistic?
A: For a categorical variable taking k possible values, if the expected values are $e_i, i = 1, \ldots, k$ and the observed values are $o_i, i = 1, \ldots, k$, then the χ^2 statistic can be computed as: $S = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{o_i}$. For the problem, since the coin is claimed to be unbiased, the expected values are $50, 50$. Further, the observed values are $54, 46$. Then, the chi-squared statistic is given by $S = \frac{(54-50)^2}{50} + \frac{(46-50)^2}{50} = 0.64$.

- For some problems, writing a short code snippet to perform some computation might be beneficial. However, you are not expected to submit the script. Note that you still have to show all key steps. For instance, a student who states that they used `numpy.mean` (or wrote a short code snippet) for the problem in Example 1 would not receive the full points.
- Do NOT change the order of the questions in your submission.
- Start each question on a new page.
- After uploading your submission, Gradescope will prompt you to select the appropriate pages for each subproblem. Make sure that you do this precisely. Unmarked or improperly marked pages *will have 5 points reduction.*
- Late submission policy: there will be a 24-hour grace period without any grade reduction, i.e., Gradescope will accept late submissions until **Thursday, April 30, 2026, at 23:59**. Unfortunately, We will NOT accept late submissions past the grace period (without a reasonable justification).

1. (24 points) Answer the following short-answer questions. Keep each response brief.

   (a) (3 points) In the K-Means algorithm, the initial placement of centroids has only a minimal impact on the quality of the resulting clustering. Discuss how initialization affects convergence and solution quality.

   (b) (3 points) What is the motivation behind using the k-medians algorithm instead of the k-means algorithm in certain situations? Is the k-medians algorithm more computationally demanding than k-means?

   (c) (3 points) Consider a clustering result and its corresponding ground-truth labels. A student claims that in a perfect clustering, the conditional entropy $H(Y \mid C)$ is maximized. Evaluate this claim and determine what value $H(Y \mid C)$ should take in the ideal case. Justify your answer.

   (d) (3 points) In the K-Means Clustering algorithm, why might a cluster center have no points assigned to it after the cluster assignment step? Explain how this situation can arise during the algorithm's iterations.

   (e) (4 points) What is the best clustering method (partition-based, hierarchical, or density) for the following task: clustering employees in a company based on their salaries and years of working experience? Justify your answer, assuming the data is continuous and free of significant outliers.

   (f) (4 points) Explain why a typical hierarchical clustering method may not lead to better quality clustering than a partitioning method.

   (g) (4 points) In DBSCAN, if a point p is **directly density-reachable** from a point q, explain why this implies that q must be a core point, but p does not necessarily have to be.

   Hint: You may use the following definitions:

   Directly density-reachable: A point p is directly density-reachable from q (w.r.t. Eps, $MinPts$) if

   $$p \in N_{Eps}(q) \quad \text{and} \quad |N_{Eps}(q)| \geq MinPts$$

   Density-reachable: A point p is density-reachable from q if there exists a chain

   $$q = p_1, p_2, \ldots, p_n = p$$

   such that each p_{i+1} is directly density-reachable from p_i.

2. (26 points) Suppose we have three clusters shown in Figure 1, containing three classes of samples ($\times$, $\circ$, $\triangle$) as follows:

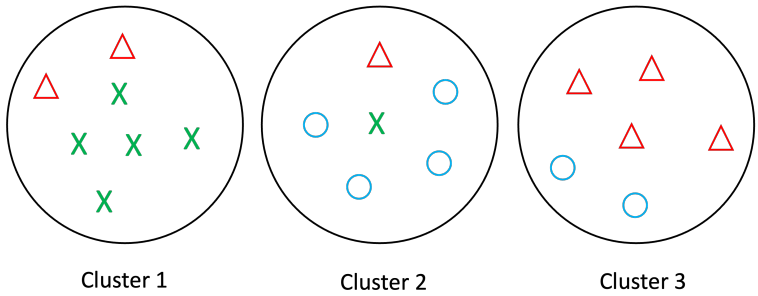

Figure 1: Three clusters with different class distributions.

cluster 1:	$\times\ \times\ \times\ \times\times,\ \triangle\triangle$	(counts: X=5, O=0, Δ=2; $\lvert C_1\rvert = 7$)
cluster 2:	$\circ\circ\circ\circ,\ \times,\ \triangle$	(counts: X=1, O=4, Δ=1; $\lvert C_2\rvert = 6$)
cluster 3:	$\triangle\triangle\triangle\triangle,\ \circ\circ$	(counts: X=0, O=2, Δ=4; $\lvert C_3\rvert = 6$)

Thus the dataset has $n = 19$ points with class totals $|X| = 6$, $|O| = 6$, $|\Delta| = 7$.

(a) (4 points) What is the **total purity** of this clustering?

(b) (6 points) What is the **precision** of each cluster?

(c) (6 points) What is the **recall** of each cluster for its majority class?

(d) (6 points) What is the **F-measure** of each cluster?

(e) (4 points) What is the **NMI** score of this clustering?
Use the natural logs.

3. (30 points) Suppose we want to cluster the following 13 points:

Index	x_1	x_2
1	1	3
2	1	2
3	2	1
4	2	2
5	2	3
6	3	2
7	5	3
8	4	3
9	4	5
10	5	4
11	5	5
12	6	4
13	6	5

Table 1: Table 3: Dataset

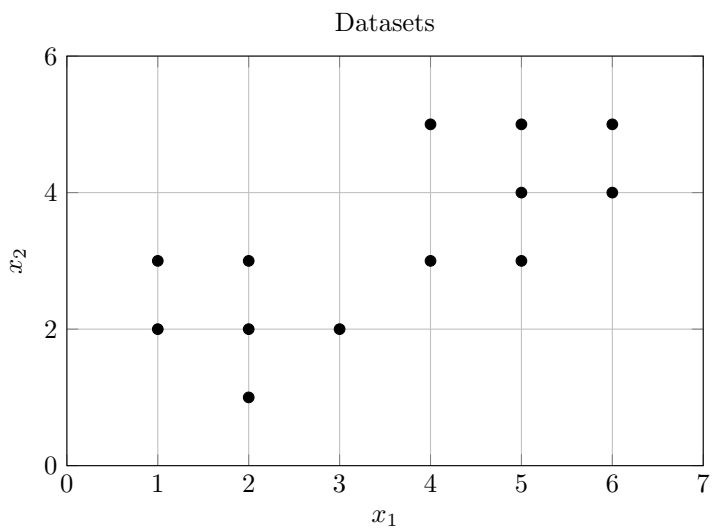

Figure 2: 2-D scatterplot of the Dataset

(a) (10 points) Cluster the above points using the K-means algorithm with $k = 2$. Please use $(0, 4)$ and $(6, 5)$ as the initial center points for the two clusters. Show your reasoning.

(b) (10 points) Now we want to use DBSCAN, a density-based algorithm, with MinPts = 2, and Eps = 1.5. Outline your clustering process.

(c) (10 points) Perform AGNES, a hierarchical clustering algorithm, on the above points. Please use the single-link method and adopt Euclidean distance as the dissimilarity measure.

4. (20 points) Consider a dataset with 5 data points a, b, c, d, e with pairwise distances given by Table 2.

	a	b	c	d	e
a	0				
b	2.1	0			
c	4	6.1	0		
d	7.8	8.6	6.1	0	
e	7.3	7.2	7.3	3.1	0

Table 2: Pairwise distance between data points. Since the distance is assumed to be symmetric, only the lower diagonal and diagonal entries are shown.

Consider running *average link* agglomerative clustering algorithm on the dataset. For *each step* of the algorithm:

(a) (10 points) Show which two clusters will be merged at step based on the pairwise distances; and

(b) (10 points) Show the updated pairwise distance matrix after merging.